VII.

On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

By G. Lejeune Dirichlet.

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On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

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According to a beautiful theorem first stated and proved by Gauss,[2] every surface can be covered with an infinitely thin mass layer whose potential at each point of the surface assumes an arbitrarily prescribed value, provided that this value varies continuously over the whole extent of the surface. The actual determination of the mass distribution which effects this, however, is at the present state of science possible only for a few special surfaces; among these, as Gauss already observed, is the whole spherical surface.

Let R denote the radius of the sphere, r the distance of any point in space from the centre, θ the angle between r and a fixed straight line, and φ the angle in the surface between the plane determined by this line and r and a fixed plane. The potential value V, prescribed on the entire surface and expressed by θ and φ, may then be expanded in the known spherical functions. Let

$$V = \sum X_n$$

be this expansion, where the summation sign extends from $n = 0$ to $n = \infty$, and let

$$\rho = \sum Y_n$$

be the analogous expansion of the density ρ to be determined. With the help of the latter, the potential v at any point in space can be represented.[3] Of the two expressions for it, one valid in the interior and the other in the exterior space, either would suffice for our purpose; the former is

$$v = 4\pi R \sum \frac{1}{2n+1} \left(\frac{r}{R}\right)^n Y_n.$$

Since this expression remains valid up to $r = R$, in which case v becomes V, and since the same function is capable of only one expansion in spherical functions, comparison with the first series

[1] The Academy report for 1850, p. 464, states: "Mr. Dirichlet gave a lecture on a new expression for the mass distribution on a spherical surface when the potential at every point of the surface is to receive an arbitrarily prescribed value." K.

[2] General theorems concerning the attractive and repulsive forces acting in inverse ratio to the square of the distance, by C. F. Gauss.

[3] Mécanique céleste, Book III, No. 13.

gives

$$Y_n = \frac{2n+1}{4\pi R} X_n,$$

and hence

$$\rho = \frac{1}{4\pi R} \sum (2n+1) X_n.$$

In an earlier paper[4] it was shown that every function prescribed arbitrarily over the whole spherical surface, that is, from $\theta = 0$, $\varphi = 0$ to $\theta = \pi$, $\varphi = 2\pi$, can be developed convergently in spherical functions, provided only that it nowhere becomes infinite. The convergence of the series for V, which is the starting point of the solution just described, is therefore beyond doubt. The matter is different for the series $\sum Y_n$ assumed for the density ρ and then determined by comparison with it. Since the existence of a completely determined mass distribution corresponding to the prescribed potential value V is compatible with the density becoming infinite at particular points or along particular lines, it remains entirely uncertain in all such cases whether the series preserves its convergence at all places where the density remains finite and whether it actually represents the density. It seemed to me not uninteresting to submit this question to closer examination, for which my earlier paper supplied the necessary aids, especially since this investigation allowed one to expect a simplification of the expression just found for ρ. That expression involves a fourfold infinite operation: a double summation and likewise a double integration. That the latter cannot be removed from the expression is clear, since the density at each point must depend on all potential values prescribed, up to continuity, over the whole extent of the surface. On the other hand, it has turned out that the density can always be represented without series by a double integral, which in many cases can even be reduced to a single integral.

1.

Put

$$V = f(\theta, \varphi).$$

It is known that for the general term X_n one has

$$X_n = \frac{2n+1}{4\pi} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi',$$

where

$$\cos\omega = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi - \varphi')$$

and where $P_n(\cos\omega)$ denotes the coefficient of α^n in the expansion of the radical expression

$$\frac{1}{\sqrt{1 - 2\alpha\cos\omega + \alpha^2}}.$$

The double integration extends from

$$\theta' = 0,\ \varphi' = 0 \qquad \text{to} \qquad \theta' = \pi,\ \varphi' = 2\pi.$$

If the divisor R is omitted, or equivalently if the radius of the sphere is set equal to unity, the general term of the series representing ρ becomes

$$\frac{(2n+1)^2}{(4\pi)^2} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi'.$$

[4] Sur les séries dont le terme général dépend de deux angles, et qui servent à exprimer des fonctions arbitraires entre des limites données. Crelle's Journal, vol. XVII. Vol. I, p. 283 of this edition of G. Lejeune Dirichlet's Werke. K.

It will plainly be enough to examine this series for the case $\theta = 0$, that is, for the pole p of the spherical polar coordinates θ, φ, since, as in the earlier paper, the result found for the point p may be transferred immediately to any other point m of the surface. Then

$$\cos\omega = \cos\theta',$$

and $P_n(\cos\omega)$ is independent of φ'. Put

$$\frac{1}{2\pi}\int_0^{2\pi} f(\theta',\varphi')\,d\varphi' = F(\theta').$$

Thus $F(\theta')$ denotes the mean value of the potential V on the circle described about p as centre with spherical radius θ'. Writing γ instead of θ', in agreement with the notation of the earlier paper to which we shall often refer, the general term becomes

$$\frac{(2n+1)^2}{8\pi}\int_0^{\pi} F(\gamma)P_n(\cos\gamma)\sin\gamma\,d\gamma.$$

If $(2n+1)^2$ is decomposed into its components $4n^2$, $4n$, 1, then the general term splits into three others, and consequently the series itself into three partial series. Since the convergence of the second and third of these series has already been shown in the earlier paper, we have only to consider the first, whose general term is

$$\frac{1}{2\pi}n^2\int_0^{\pi} F(\gamma)P_n(\cos\gamma)\sin\gamma\,d\gamma.$$

Taking the sum of the first $n+1$ terms, expressing $P_n(\cos\gamma)$ by means of a definite integral according to equation (3) of the earlier paper,[5] and reversing the order of the two integrations, one obtains

$$\frac{1}{\pi^2}\int_0^{\pi}(\cos\psi + 4\cos 2\psi + \cdots + n^2\cos n\psi)\Pi(\psi)\,d\psi,$$

where, as before,

$$\Pi(\psi) = \sin\frac{\psi}{2}\int_0^{\psi} F(\gamma)\sin\gamma\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\int_{\psi}^{\pi} F(\gamma)\sin\gamma\frac{d\gamma}{E},$$

$$\Delta = \sqrt{2(\cos\gamma - \cos\psi)}, \qquad E = \sqrt{2(\cos\psi - \cos\gamma)}.$$

Since the integrals contained in $\Pi(\psi)$ are functions of ψ which remain continuous from $\psi = 0$ to $\psi = \pi$ (by §3 of the earlier paper), the same property belongs to the function $\Pi(\psi)$ itself.

The present investigation also requires discussion of the derivative $\Pi'(\psi)$. In forming it, because of the denominators Δ and E contained in the integrals, a preliminary transformation by integration by parts is necessary. Taking into account that $F(\gamma)$, from its origin as the continuous potential value $f(\theta,\varphi)$, is itself continuous, this double operation gives

$$\begin{aligned}\Pi'(\psi) = (F(0) - F(\pi))\sin\psi + \frac{1}{2}\cos\frac{\psi}{2}\int_0^{\psi} F'(\gamma)\Delta\,d\gamma + \frac{1}{2}\sin\frac{\psi}{2}\int_{\psi}^{\pi} F'(\gamma)E\,d\gamma \\ + \sin\frac{\psi}{2}\sin\psi\int_0^{\psi} F'(\gamma)\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\sin\psi\int_{\psi}^{\pi} F'(\gamma)\frac{d\gamma}{E}.\end{aligned}$$

We now assume that $F'(\gamma)$ remains finite everywhere from $\gamma = 0$ to $\gamma = \pi$. Then all components of $\Pi'(\psi)$, and hence $\Pi'(\psi)$ itself, will be finite and continuous from $\psi = 0$ to $\psi = \pi$. For the first three terms this is evident; for the last two, arguments very similar to those cited above show that the integrals contained in them retain this double property so long as, in the first, $\psi < \pi$

[5] Vol. I, p. 291 of this edition of G. Lejeune Dirichlet's Werke. K.

and, in the second, $\psi > 0$ is assumed. For $\psi = \pi$ and $\psi = 0$ these integrals may respectively take infinite values, but, as is easy to see, their order cannot exceed

$$\log\left(\frac{1}{\cos\frac{\psi}{2}}\right) \qquad \text{and} \qquad \log\left(\frac{1}{\sin\frac{\psi}{2}}\right),$$

so that the terms themselves become zero.

For what follows one must also know the second derivative $\Pi''(\psi)$ for the special value $\psi = 0$. Since $\Pi'(0) = 0$, the desired value is most easily found by letting ψ become infinitely small in the quotient $\frac{1}{\psi}\Pi'(\psi)$. Thus one obtains

$$\Pi''(0) = F(0) - F(\pi) + \frac{1}{2}\int_0^\pi F'(\gamma)\sin\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma,$$

or, if the third term is integrated by parts,

$$\Pi''(0) = F(0) - \frac{1}{2}F(\pi) - \frac{1}{4}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.$$

If, in addition to the finiteness of $F'(\gamma)$ just assumed and the consequent continuity of $\Pi'(\psi)$, a definite finite value also exists for $\Pi''(0)$, or what is the same, if the second integral has such a value, then our series will always converge and its sum will be easy to state. Indeed, for shortness put

$$X(\psi) = \frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}} = \frac{1}{2} + \cos\psi + cos 2\psi + \cdots + \cos n\psi.$$

The above expression for the sum of the first $n+1$ terms is then

$$-\frac{1}{\pi^2}\int_0^\pi \Pi(\psi)X''(\psi)\,d\psi.$$

Since the boundary term emerging after twice integrating by parts,

$$\Pi(\psi)X'(\psi) - \Pi'(\psi)X(\psi),$$

is continuous and vanishes at both limits, this operation gives

$$-\frac{1}{\pi^2}\int_0^\pi \Pi''(\psi)\frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}}\,d\psi.$$

By a known theorem, this expression tends, as n increases, even when $\Pi''(\psi)$ becomes infinite for one or more non-zero values of ψ, to the limit

$$\begin{aligned} -\frac{1}{2\pi}\Pi''(0) = &-\frac{1}{2\pi}F(0) + \frac{1}{4\pi}F(\pi) + \frac{1}{8\pi}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma \\ &- \frac{1}{4\pi}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma. \end{aligned}$$

Adding to this value those of the two other series, as found in the earlier paper, gives for the density at p

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma\right].$$

It remains to examine how the series expressing the density behaves, as regards convergence, in the case where the integral

$$\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma$$

and therefore also the expression just found for ρ retains a definite finite value, while the function $F'(\gamma)$ becomes infinite for one or more non-zero values of γ. According to the above, the condition for convergence then consists solely in this: the function $\Pi''(\psi)$ derived from $F'(\gamma)$ must remain finite and continuous from $\psi = 0$ to $\psi = \pi$. A more exact consideration of the way $\Pi''(\psi)$ is formed shows that, if the becoming infinite of $F'(\gamma)$ occurs in such a way that for every value c at which $F'(\gamma)$ takes an infinitely large value, $\sqrt{\varepsilon}F'(c \pm \varepsilon)$ is itself infinitely small for infinitely small ε, then the continuity of $\Pi''(\psi)$ obtains just as in the previously examined case of an everywhere finite function $F'(\gamma)$, and with it the convergence of the series representing the density. In contrast, in general divergence occurs when the condition just stated is no longer fulfilled. Although the proof of this result presents no essential difficulty, it would require too much space and would be of too little interest to carry it out here. For the secure use of the series it is enough to know, from the preceding discussion, the only cases in which the convergence of the series can cease. Moreover, an occasion will arise below to demonstrate the divergence actually occurring in such a case by means of a very simple example.

2.

Since the preceding derivation of the finite expression found for ρ loses its validity when the series ceases to converge, a proof of this expression applicable to all cases is still needed.

According to a known theorem, the derivative $\frac{\partial v}{\partial r}$, when θ and φ are kept constant and r is allowed to approach unity, tends to two different limits according as $r < 1$ or $r > 1$ is assumed throughout. If these limiting values are denoted respectively by K and L, then the density at the corresponding point of the surface is determined by

$$\rho = \frac{1}{4\pi}(K - L).$$

Between the limits K, L and the potential value V on the surface there is a simple relation, so that only one of the two limiting values has to be found. Namely, if v and v_1 are the potential values for two points on the same radius vector, their distances from the centre, r and r_1, satisfying

$$rr_1 = 1,$$

then, as is readily seen,

$$v_1 = \frac{1}{r_1}v,$$

and therefore

$$\frac{\partial v_1}{\partial r_1} = -\frac{1}{r_1^2}v + \frac{1}{r_1}\frac{\partial v}{\partial r}\frac{\partial r}{\partial r_1} = -\frac{1}{r_1^2}v - \frac{1}{r_1^3}\frac{\partial v}{\partial r}.$$

Thus one sees that

$$L = -V - K$$

and

$$\rho = \frac{1}{4\pi}(V + 2K).$$

To determine K it suffices to use the expression for v valid in the interior space:

$$v = \frac{1 - r^2}{4\pi} \iint \frac{f(\theta', \varphi') \sin\theta' \, d\theta' \, d\varphi'}{(1 - 2r\cos\omega + r^2)^{3/2}}.$$

This is easily proved without developing in series. It is enough to observe that this expression, as r approaches its upper limit 1, passes by a known and often treated theorem into $f(\theta, \varphi) = V$, and that on the other hand it satisfies the partial differential equation first set up by Laplace for v, since

$$\frac{1-r^2}{(1-2r\cos\omega+r^2)^{3/2}},$$

considered as a function of the polar coordinates r, θ, φ, is a particular integral of this equation. In the case $\theta = 0$, the expression takes, writing γ again instead of θ' and keeping the earlier notation, the simple form

$$v = \frac{1}{2}(1-r^2)\int_0^\pi \frac{F(\gamma)\sin\gamma\, d\gamma}{(1-2r\cos\gamma+r^2)^{3/2}}.$$

If one differentiated the expression in this form, the limiting value of $\frac{\partial v}{\partial r}$ would be difficult to determine. It is therefore expedient first to perform an integration by parts. One obtains

$$2v = \left(\frac{1}{r}+1\right)F(0) - \left(\frac{1}{r}-1\right)F(\pi) + \left(\frac{1}{r}-r\right)\int_0^\pi \frac{F'(\gamma)\, d\gamma}{(1-2r\cos\gamma+r^2)^{1/2}},$$

where passage to the limit after differentiation now presents no further difficulty and gives

$$K = -\frac{1}{2}F(0) + \frac{1}{2}F(\pi) - \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\, d\gamma.$$

Taking into account that for the point p,

$$V = F(0),$$

it follows that

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\, d\gamma\right],$$

which agrees with the formula derived from the series.

3.

Transferring the result found for the point corresponding to $\theta = 0$ to an arbitrary point m, one obtains the following general rule for determining the density.

By a first integration, find for the circle described on the surface about m as centre with arbitrary spherical radius λ the mean value of the given potential V. Denote this mean by $\varphi(\lambda)$. Then the density sought is given by

$$\rho = \frac{1}{4\pi}\left[\varphi(\pi) - \int_0^\pi \frac{\varphi'(\lambda)}{\sin\frac{\lambda}{2}}\, d\lambda\right].$$

One may also remark that, by the meaning of the function $\varphi(\lambda)$, the special value $\varphi(\pi)$ denotes the potential value at the point antipodal to m. It hardly needs saying that the function $\varphi(\lambda)$ will be different for every point m, or, in other words, that besides λ it will also contain the two quantities which determine the position of m on the surface.

It is perhaps not superfluous, however, to forestall an error which can easily arise from an inattentive consideration of the result just stated. At first sight, because the denominator under the integral sign vanishes at the lower limit, one is tempted to suppose that the integral remains finite only for special positions of the point m, but in general becomes infinite; in fact, precisely the reverse is the case. First, by the continuity of $\varphi(\lambda)$ it is easy to see that the part of the integral extending from any positive value δ, however small, to the upper limit π always remains determinate and finite, however often $\varphi'(\lambda)$ itself may become infinite in this interval. That the same is also true, apart from singular cases, for the part extending from 0 to δ has its reason in the fact that for small values of λ, $\varphi'(\lambda)$ is generally of the same first order as the denominator. If the potential value V is represented as a function $\chi(\lambda, \psi)$ of spherical polar coordinates λ, ψ whose pole is the point m, so that

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\lambda, \psi)\, d\psi,$$

then the property just mentioned does not appear only because $\chi(\lambda, \psi)$ is not arbitrary, but must satisfy the special conditions of being periodic with respect to ψ and becoming independent of ψ for $\lambda = 0$. One is immediately convinced of the assertion made above if one projects the spherical surface onto an arbitrary plane, whose points are referred to two rectangular axes t, u, and then expresses V at each point by the coordinates t, u of its projection. This representation has the disadvantage that, in order to cover the whole surface, two functions of t and u must be considered; but this disadvantage disappears for our purpose, which requires consideration only of an arbitrarily small surface element containing the point m. Taking the tangent plane at m as the plane of projection, m as the origin of coordinates, and putting $V = \chi(t, u)$, this new function has to satisfy no condition other than continuity. Since evidently one may take

$$t = \sin\lambda \cos\psi, \qquad u = \sin\lambda \sin\psi,$$

the function $\varphi(\lambda)$ assumes the form

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\sin\lambda \cos\psi, \sin\lambda \sin\psi)\, d\psi.$$

In this form the occurrence of the above property is evident, provided the partial derivatives of $\chi(t, u)$ up to and including the second order, for the case where $t = 0$ and $u = 0$ simultaneously, retain definite finite values. Moreover, for the finiteness of ρ it is not even necessary that $\varphi'(\lambda)$ be of the first order for small values of the variable; it is enough that the order of $\varphi'(\lambda)$ agree with that of some positive power of λ.

As with the becoming infinite of ρ and the consequent divergence of the series, so also with the case where the series ceases to converge while the density remains finite. This case is even more restricted than the one just discussed, as is easily seen by closer consideration of the conditions on which it depends.

4.

In order to illustrate the actual occurrence of divergence of the series for ρ in the exceptional cases indicated above by a simple example, suppose that the potential value V does not contain the angle φ and is represented by $f(\theta)$. Then, as is known, the spherical-function series take a simpler form, and one has

$$V = \frac{1}{2} \sum (2n+1) A_n P_n(\cos\theta), \qquad \rho = \frac{1}{8\pi} \sum (2n+1)^2 A_n P_n(\cos\theta),$$

where the general coefficient A_n is given by

$$A_n = \int_0^{\pi} f(\theta) P_n(\cos\theta) \sin\theta\, d\theta.$$

Put

$$f(\theta) = \sqrt{\cos\theta} \quad \text{as long as } 0 < \theta < \frac{1}{2}\pi, \qquad f(\theta) = 0 \quad \text{when } \theta \text{ lies between } \frac{1}{2}\pi \text{ and } \pi.$$

Then the requirement of continuity is satisfied, and the integral

$$A_n = \int_0^{\frac{1}{2}\pi} P_n(\cos\theta)\sqrt{\cos\theta}\sin\theta\,d\theta$$

has to be determined. Since its evaluation by means of the known expressions for $P_n(\cos\theta)$ does not immediately give the result in the simplest form, the following way is preferable.

From

$$\sum P_n(\cos\theta)\alpha^n = \frac{1}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

the required integral is the coefficient of α^n in the expansion of

$$\int_0^{\frac{1}{2}\pi} \frac{\sqrt{\cos\theta}\sin\theta\,d\theta}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

where the proper fraction α may be regarded as positive. By the substitution $t=\sqrt{\cos\theta}$ and by carrying out the integration, one obtains

$$2\int_0^1 \frac{t^2\,dt}{\sqrt{1-2\alpha t^2+\alpha^2}} = \sqrt{\frac{1}{8\alpha^3}}\left((1+\alpha^2)\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} - (1-\alpha)\sqrt{2\alpha}\right).$$

For convenience in developing the expression, put

$$z = \arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}}.$$

Then differentiating gives

$$\frac{dz}{d\alpha} = \frac{1+\alpha}{1+\alpha^2}\sqrt{\frac{1}{2\alpha}} = \frac{1}{\sqrt{2}}(\alpha^{-\frac{1}{2}} + \alpha^{\frac{1}{2}} - \alpha^{\frac{3}{2}} - \alpha^{\frac{5}{2}} + \cdots),$$

and, on integrating and observing that z vanishes with α,

$$\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} = \sqrt{2}\left(\alpha^{\frac{1}{2}} + \frac{1}{3}\alpha^{\frac{3}{2}} - \frac{1}{5}\alpha^{\frac{5}{2}} - \frac{1}{7}\alpha^{\frac{7}{2}} + \cdots\right).$$

Substitution gives for the required expansion

$$-\frac{1}{2}\left(\left(\frac{1}{-1}-\frac{1}{3}\right) - \left(\frac{1}{1}-\frac{1}{5}\right)\alpha - \left(\frac{1}{3}-\frac{1}{7}\right)\alpha^2 + \left(\frac{1}{5}-\frac{1}{9}\right)\alpha^3 + \cdots\right),$$

where the signs inside the parentheses form the four-term period

$$+\ -\ -\ +,$$

which may be represented by $(-1)^{\frac{1}{2}n(n+1)}$. Thus

$$A_n = -(-1)^{\frac{1}{2}n(n+1)}\frac{2}{(2n-1)(2n+3)},$$

and the preceding series take the form

$$V = -\sum(-1)^{\frac{1}{2}n(n+1)}\frac{2n+1}{(2n-1)(2n+3)}P_n(\cos\theta),$$

$$\rho = -\frac{1}{4\pi}\sum(-1)^{\frac{1}{2}n(n+1)}\frac{(2n+1)^2}{(2n-1)(2n+3)}P_n(\cos\theta).$$

From the remarks in the preceding article, one sees without difficulty in the present case that the derivative $\varphi'(\lambda)$ becomes infinite in such a way that the condition stated at the end of Article 1 is not fulfilled only when $\lambda = \frac{1}{2}\pi$ and the point m corresponds to $\theta = 0$ or $\theta = \pi$; in these cases, however, $\varphi'(\lambda)$ is of order λ for small values of λ, and ρ therefore retains a definite finite value. One also sees that the density becomes infinite only at the equator, or for $\theta = \frac{1}{2}\pi$, since then $\varphi'(\lambda)$ is of order λ^{-1} for small λ. In these three cases the series therefore ceases to converge. This is immediately verified if one takes into account that $P_n(\cos\theta)$ has the values 1 and $(-1)^n$ respectively for $\theta = 0$ and $\theta = \pi$, while for $\theta = \frac{1}{2}\pi$ it vanishes when n is odd and, when n is even, equals

$$\frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n} (-1)^{\frac{n}{2}},$$

which, as is known, is of order $n^{-\frac{1}{2}}$ for large n.

In the first two cases, where ρ retains a definite finite value, the series has the character of an oscillating one: among every four consecutive terms, two have positive and two negative sign, and their terms tend to unity as a limit. In the third case, all terms have the same sign and yield an infinitely large sum.

5.

Although the method by which we have just determined the coefficient A_n is no longer applicable if, while retaining the assumption $f(\theta) = 0$ for the second part of the interval, one takes $f(\theta) = \cos^k\theta$ in the first part, where k denotes an arbitrary positive constant, the very simple form of the result found for the special value $k = \frac{1}{2}$ leads one to expect something similar in the more general case. Since the simplest expression for A_n is somewhat hidden, we pause briefly to find it. Poisson, who in one of his papers[6] chose precisely the function just defined as an example of development in spherical functions, gives the coefficient A_n the form

$$A_n = \frac{1 \cdot 3 \cdots (2n-1)}{1 \cdot 2 \cdots n} \left(\frac{1}{k+n+1} - \frac{n(n-1)}{2(2n-1)} \frac{1}{k+n-1} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} \frac{1}{k+n-3} - \cdots \right).$$

This is the form in which the coefficient is obtained directly when the usual expression for $P_n(\cos\theta)$ is used in the expansion. It does not reveal the considerable simplification of which it is capable, and that simplification would not easily be suspected unless one were led to it by a special case such as the one above. The following very short, though not elementary, route leads to the simplest expression for A_n.

By §1 of the earlier paper,[7] the two integrals

$$\frac{2}{\pi} \int_0^\gamma \frac{\cos n\psi \cos\frac{\psi}{2}}{\sqrt{2(\cos\psi - \cos\gamma)}}\, d\psi, \qquad -\frac{2}{\pi} \int_0^\gamma \frac{\sin n\psi \sin\frac{\psi}{2}}{\sqrt{2(\cos\psi - \cos\gamma)}}\, d\psi$$

are respectively the coefficients of α^n in the expansions of

$$\frac{1}{2} + \frac{1}{2} \frac{1+\alpha}{\sqrt{1 - 2\alpha\cos\gamma + \alpha^2}}, \qquad -\frac{1}{2} + \frac{1}{2} \frac{1-\alpha}{\sqrt{1 - 2\alpha\cos\gamma + \alpha^2}}.$$

Adding gives

$$P_n(\cos\gamma) = \frac{2}{\pi} \int_0^\gamma \frac{\cos(n+\frac{1}{2})\psi}{\sqrt{2(\cos\psi - \cos\gamma)}}\, d\psi.$$

[6] Connaissance des temps pour l'an 1829.

[7] Vol. I, pp. 292–294 of this edition of G. Lejeune Dirichlet's Werke. K.

Multiplying this equation by $\cos^k\gamma\sin\gamma\,d\gamma$, then integrating from $\gamma=0$ to $\gamma=\frac{1}{2}\pi$ and reversing the order of integrations on the right, yields

$$A_n=\frac{2}{\pi}\int_0^{\frac{1}{2}\pi}d\psi\,\cos(n+\tfrac{1}{2})\psi\int_\psi^{\frac{1}{2}\pi}\frac{\cos^k\gamma\sin\gamma}{\sqrt{2(\cos\psi-\cos\gamma)}}\,d\gamma.$$

If one introduces a new variable t by $\cos\gamma=t\cos\psi$, the two integrations separate:

$$A_n=\frac{\sqrt{2}}{\pi}\int_0^1\frac{t^k}{\sqrt{1-t}}\,dt\cdot\int_0^{\frac{1}{2}\pi}\cos^{k+\frac{1}{2}}\psi\cos(n+\tfrac{1}{2})\psi\,d\psi.$$

Now, by a known formula,

$$\int_0^{\frac{1}{2}\pi}\cos^{p-1}\psi\cos q\psi\,d\psi=\frac{\Gamma(p)}{\Gamma\left(\frac{1}{2}(1+p+q)\right)\Gamma\left(\frac{1}{2}(1+p-q)\right)}\cdot\frac{\pi}{2^p},$$

where the constant p must be positive, and the transcendental function $\Gamma(l)$, when the argument l becomes negative, is to be understood by the relation $\Gamma(l+1)=l\Gamma(l)$, or, what is the same thing, by Gauss's definition, which comprises positive and negative arguments uniformly. By this formula and by the known representation of a Eulerian integral of the first kind by three of the second, our equation becomes

$$A_n=\frac{\Gamma(k+1)}{\Gamma\left(\frac{1}{2}(k+n+3)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)}\cdot\frac{\sqrt{\pi}}{2^{k+1}},$$

or

$$A_n=\frac{k(k-1)(k-2)\cdots(k-n+1)}{\frac{1}{2}(k+n+1)\left(\frac{1}{2}(k+n+1)-1\right)\cdots\left(\frac{1}{2}(k+n+1)-n\right)}\cdot\frac{\Gamma(k-n+1)}{\Gamma\left(\frac{1}{2}(k-n+1)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)}\cdot\frac{\sqrt{\pi}}{2^{k+1}}.$$

If for the second factor one substitutes its known value $2^{k-n}\pi^{-\frac{1}{2}}$, this gives

$$A_n=\frac{k(k-1)(k-2)\cdots(k-n+1)}{(k+n+1)(k+n-1)(k+n-3)\cdots(k-n+1)},$$

where the factors in the denominator decrease by two units.

It hardly needs mention that this simple expression can be verified with the greatest ease. For instance, it may be decomposed into partial fractions in order to recognize its agreement with the expression cited above. If one does not wish to use the latter for this verification, one may use instead the equation

$$(n+1)A_{n+1,k}=(2n+1)A_{n,k+1}-nA_{n-1,k},$$

which follows immediately from the relation between every three consecutive coefficients in the expansion of $P_n(\cos\gamma)$.

To simplify the expression still further, one distinguishes whether n is even or odd and obtains respectively

$$A_n=\frac{k(k-2)\cdots(k-n+2)}{(k+1)(k+3)\cdots(k+n+1)},\qquad A_n=\frac{(k-1)(k-3)\cdots(k-n+2)}{(k+2)(k+4)\cdots(k+n+1)}.$$

With the help of these expressions and the known formulae which give the approximate values of factorials containing a very large number of factors, one may easily verify that the special case $k=\frac{1}{2}$ examined above is exactly the limiting case for the divergence of the series occurring at $\theta=0$ and $\theta=\pi$, and that the divergence ceases as soon as $k>\frac{1}{2}$. The situation is different for the value $\theta=\frac{1}{2}\pi$: at that place the divergence and the infinitely large value of the density persist as long as $k>1$ is not assumed.

6.

We shall now make one application of the finite expression found for the density to a special case. Let V again be regarded as a mere function of θ, which may be put in the form $f(\cos\theta)$. Then ρ too depends only on the spherical distance θ of the point m from the pole p. On the circle described about m as centre with spherical radius λ, the fundamental formula of trigonometry gives

$$V = f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi),$$

where ψ denotes the angle between an arbitrary radius λ and the one directed from m to p. Since this expression has the same value for ψ and $-\psi$, the integral expressing the mean value $\varphi(\lambda)$ may be taken from $\psi = 0$ to $\psi = \pi$ and then doubled. Thus

$$\varphi(\lambda) = \frac{1}{\pi}\int_0^\pi f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi.$$

Whenever $\varphi'(\lambda)$ can be represented without an integral sign, which is especially the case when $f(z)$, or more generally $f'(z)$, is a rational function of z, whether the form of that function remains the same throughout the whole interval between $z = -1$ and $z = +1$ or changes in places without violating continuity, ρ can be reduced to a single quadrature or even expressed without any integral sign.[8]

Among the cases belonging here we consider only the one in which

$$f(\cos\theta) = \cos\theta$$

as long as $\theta < \frac{1}{2}\pi$, and

$$f(\cos\theta) = 0$$

when $\theta > \frac{1}{2}\pi$. The integral then extends only over that part of the interval between $\psi = 0$ and $\psi = \pi$ in which

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi$$

assumes positive values. Suppose first that $\theta < \frac{1}{2}\pi$; the case $\theta > \frac{1}{2}\pi$ is easily reduced to this. A simple discussion of the preceding expression shows that, so long as λ is less than $\frac{1}{2}\pi - \theta$, the integral is to be taken from $\psi = 0$ to $\psi = \pi$; for values of λ between $\frac{1}{2}\pi - \theta$ and $\frac{1}{2}\pi + \theta$, the limits of the integral are 0 and the angle ψ_1 lying between 0 and π determined by

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi_1 = 0;$$

finally, for $\lambda > \frac{1}{2}\pi + \theta$, the integral vanishes, since the preceding expression is then always negative. Thus, in the three intervals just distinguished, the function $\varphi(\lambda)$ is

$$\cos\theta\cos\lambda, \qquad \frac{1}{\pi}\int_0^{\psi_1}(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi, \qquad 0.$$

In the second case the actual integration is postponed until after differentiation, in order to shorten the calculation. On differentiating, the term arising from the variability of the upper limit ψ_1 vanishes by the equation determining ψ_1, and one obtains the following three expressions for $\varphi'(\lambda)$:

$$-\cos\theta\sin\lambda, \qquad \frac{1}{\pi}(-\psi_1\cos\theta\sin\lambda + \sin\psi_1\sin\theta\cos\lambda), \qquad 0.$$

[8] In the general case, where V contains both angles θ, φ, the reduction of ρ to a single quadrature always takes place if, putting $x = \cos\theta$, $y = \sin\theta\cos\varphi$, $z = \sin\theta\sin\varphi$, one can represent V by a function $f(x, y, z)$ of the quantities x, y, z whose three first partial derivatives are rational, integral or fractional functions of x, y, z; again the form of $f(x, y, z)$ may be different in different parts of the surface. This is a result remarkable for its great range.

These formulas also show in passing that, at least so long as m remains on the hemisphere for which they apply, $\varphi'(\lambda)$ does not become infinite and, for a small λ for which the first formula holds, remains of the first order. In this respect there is an exception only for $\theta = \frac{1}{2}\pi$: then the two outer intervals disappear and $\varphi'(\lambda)$ everywhere has the value $\frac{1}{\pi}\cos\lambda$, which for small values of λ is of order zero; hence an infinitely large density occurs at the equator.

According to the expressions found for $\varphi'(\lambda)$, the integral contained in the general expression for ρ splits into two others, extending respectively from 0 to $\frac{1}{2}\pi - \theta$ and from $\frac{1}{2}\pi - \theta$ to $\frac{1}{2}\pi + \theta$. The first has the simple value

$$-4\cos\theta \sin\left(\frac{1}{4}\pi - \frac{1}{2}\theta\right).$$

The second, consisting of two parts, is, before taking the limits into account,

$$\frac{\sin\theta}{\pi}\int \frac{\sin\psi_1 \cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda - \frac{2\cos\theta}{\pi}\int \psi_1 \cos\frac{1}{2}\lambda\,d\lambda.$$

Integration by parts in the last term gives it the form

$$\frac{\sin\theta}{\pi}\int \frac{\sin\psi_1 \cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda + \frac{4\cos\theta}{\pi}\int \frac{\partial\psi_1}{\partial\lambda}\sin\frac{1}{2}\lambda\,d\lambda - \frac{4}{\pi}\psi_1\cos\theta\sin\frac{1}{2}\lambda.$$

Since the equation determining ψ_1 gives, at the two limits, respectively $\psi_1 = \pi$ and $\psi_1 = 0$, the term free of the integral sign contributes, on passing to the limits, a value opposite to that found above for the first integral. Thus only the first two terms remain. Substitution of the expressions obtained from the equation for ψ_1,

$$\sin\psi_1 = \frac{\sqrt{\sin^2\theta - \cos^2\lambda}}{\sin\theta\sin\lambda}, \qquad \frac{\partial\psi_1}{\partial\lambda} = -\frac{\cos\theta}{\sin\lambda}\frac{1}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

gives them the form

$$\frac{1}{\pi}\int \frac{\cos\lambda}{\sin\lambda\sin\frac{1}{2}\lambda}\sqrt{\sin^2\theta - \cos^2\lambda}\,d\lambda - \frac{4\cos^2\theta}{\pi}\int \frac{\sin\frac{1}{2}\lambda}{\sin\lambda}\frac{d\lambda}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

where the integrals are to be taken from $\lambda = \frac{1}{2}\pi - \theta$ to $\lambda = \frac{1}{2}\pi + \theta$. Introducing as new variable

$$z = \frac{\cos\lambda}{\sin\theta},$$

the limits for it become $+1$ and -1; changing the sign, they become -1 and $+1$. Substitution in the expression for ρ, while also taking account of the vanishing of $\varphi(\pi)$, gives

$$\rho = \frac{1}{2\sqrt{2}\pi^2}\int_{-1}^{+1}\left(\left(\frac{2 - z\sin\theta}{1 - z^2\sin^2\theta}\right)\cos^2\theta - z\sin\theta\right)\frac{dz}{\sqrt{(1-z^2)(1-z\sin\theta)}}.$$

This result only appears to contain a component depending on elliptic integrals of the third kind. Since the limits are two consecutive values for which the radical vanishes, the expression, brought to the usual form, will contain only so-called complete elliptic integrals[9] and hence, by a beautiful theorem due to Legendre, can be reduced to the first and second kinds.

Finally, let us note how the case of the problem treated in the paper mentioned above, for a surface which differs only slightly from a spherical surface, can be reduced to the case of the sphere without a series expansion.

Let

$$r = 1 + \gamma z$$

[9] Fundamenta nova theoriae functionum ellipticarum, by C. G. J. Jacobi, p. 14. C. G. J. Jacobi's collected works, vol. I, p. 67. K.

be the equation of the surface, where γ is a small constant whose higher powers are to be neglected, and z denotes a function of θ and φ. Agree to denote by an added accent the value which a function of θ, φ takes when θ, φ are replaced by θ', φ'. Let ds' be the element of the surface belonging to θ', φ', and let p be the distance between the two points of the surface corresponding to the coordinate pairs θ, φ and θ', φ'. The problem requires that the equation

$$V = \int \rho' \frac{ds'}{p}$$

be satisfied for all values of θ, φ in the same range, the double integration extending over the whole surface, or from $\theta' = 0$, $\varphi' = 0$ to $\theta' = \pi$, $\varphi' = 2\pi$. If $d\sigma'$ is the element of the spherical surface corresponding to ds', cut off by the same radius vector as ds', and if q is the distance between the points on the spherical surface belonging to θ, φ and θ', φ', then the simplest geometrical considerations immediately give

$$ds' = (1 + 2\gamma z') d\sigma'$$

and

$$p = q\left(1 + \frac{1}{2}\gamma(z + z')\right).$$

Substitution of these expressions turns our equation into

$$V = \int \rho' \left(1 + \gamma\left(\frac{3}{2}z' - \frac{1}{2}z\right)\right) \frac{d\sigma'}{q},$$

or

$$V + \frac{1}{2}\gamma z \int \rho' \frac{d\sigma'}{q} = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right) \frac{d\sigma'}{q}.$$

Since, by the penultimate equation, the integral multiplied by γ in the last equation differs from V only by a quantity of order γ, V may be put for it. One obtains

$$\left(1 + \frac{1}{2}\gamma z\right) V = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right) \frac{d\sigma'}{q},$$

which reduces the problem to the case of the spherical surface. Thus one sees that the given potential value, multiplied by $1 + \frac{1}{2}\gamma z$, is to be regarded as applying to the spherical surface, and that the density determined on this assumption is then to be divided by $1 + \frac{3}{2}\gamma z$.